L2.4 The echelon form

Transcript Language

English

Hello and welcome to the maths 2 component of the online BSc course on Data Science.

In today's video we are going to talk about the echelon form. So, the Echelon form is

a particular form for a matrix and when the matrix is in this form, we can read off solutions

quite easily.

So, let us look at, let us recall first what is the system of linear equations. So, a general

system of m linear equations within n unknowns is like this a 1 1 x 1 plus a 1 2 x 2 plus

a 1 n x n is b 1 and then all the way up to a m 1 x 1 plus a m 2 x 2 plus a m n x n is

bm. So, there are n unknowns, which is the xi s and there are m equations. So, for each

equation we have a constant on the right-hand side. So, the i th equation that constant

is bi.

We have seen earlier that, we can write this in a vector form in a matrix form that is,

so we consider the m by n matrix A, which is a matrix of coefficients and x is the vector

of unknowns x1, x2, xn and b is the vector of constants, the column matrix b1, b2, bm

And then what is the solution?

So, solution is an assignment of values for x so that the equations are satisfied. So,

when you put xi is equal to some particular number and for each xi I put in a number and when the equation holds true, that means, indeed when you compute a 1 1 times whatever

value you have substituted for x1 plus a 1 2 times whatever value of substitute for x2

plus a 1 n times whatever value you have substituted for xn is b1 and the same thing happens for

all the other equations, then we say that those particular values that we have substituted

constitute a solution for x1, x2, xn.

So, let us look at the example of this matrix. So, we have A is the matrix with first row

1,0,2, and second row 0,1,3. So, let us see how we can find out solutions for this. So,

here, I am going to write down what the corresponding equations are, of course, here I have not

mentioned what the constants are. So, let us take the equation A x equals 0. So, this

is called a homogeneous system, which we will study later. So, the corresponding equations

are going to be x1 plus 2 x3 is 0 and x2 plus 3 x3 is 0.

So, now, typically, what do we do, we try to manipulate around and find values for x1, x2, x3. If you look at this, this is not really much we can do to manipulate this and the

reason is that we can have many, many solutions. So, how do we find solutions? So, you can

see from these equations that x1 is minus 2 x3 and x2 is minus 3 x3 and that is all

that we need, in order to get a solution. So, whenever we have, we substitute a value

for x1, which is negative 2 times the value for x3 and x2, which is negative 3 times the

value for x3, we get a solution.

So, you can put x3 equals some constant c and then let us say I take x3 equals 5. So,

if x3 is 5, then to get my solution, what I can do is I can put x1 is minus 10 and x2

is minus 15 and indeed, the one possible solution then is x is equal to minus 10, minus 15,

and 5, so 5 was an arbitrary choice for x3, I could have used some other number. So, for

example, if I use x3 is c, then I get x1 is minus 2c and x2 is minus 3c.

So really, the set of solutions for this, we can write this as minus 2c, minus 3c, and c where c can be any real number. So, we can have c is 5, c is 20, c is 1 million, whatever

you want and all of these, for each value of c we will get a solution. So, we have infinitely

many solutions, that is the first thing we observed and not only that, we in fact see

that we can get these all these solutions in a very particular way, we can really do

this for any b.

So, if you carefully observe what we did, instead of Ax is 0, if we had Ax is b, we

can obtain solutions for Ax is b as well, maybe I will describe this very fast. So,

for Ax is b, what do we do, for Ax is b, what, what we do is we write down again, the same

set of equations, the equations are x1 plus 2 x3 is b1 and x2 plus 3 x3 is b2. And from

here, we can move x3 to the other side and then whatever value of x3 we have, we can

read off the values of x1 and x2 and each time we will get a solution. So again, we

have an infinite number of solutions.

So, there was something particularly easy about this matrix, we could read off all the

solutions not only to Ax is 0, but Ax is b any b. So, matrices of this form, have a particular,

there is a name for this form, it is called the Row echelon form and that is what we are

going to study in this video.

So, when it is said to be in Row echelon form. So, it is said to be in Row echelon form,

if the first non-zero element in each row called the leading entry is 1. So, whenever

you have a row and it is a non-zero row, so you could have 0 rows of course, if it is

a non-zero row, then there will be a first element which is non-zero as you move from

left to right. And the first such 1 has to be 1.

The next axiom for the Row echelon form is that each leading entry is in a column, which

is to the right of the leading entry in the previous row. So, like you saw in the previous

example, in the first row, the 1 was in the 1 1 th place, in the second row, the 1 was in the 2 2 th place. So, if you consider the second column and you ask, or rather, you

consider the second row and you ask where is the leading entry, which is the first 1

that you have? It is in the second column and that is to the right of the leading entry

in the previous column, which was in the first row.

So, broadly speaking, you have to go downwards and to the right, like this, the 1s have to

be like this. Let us look at the next requirement. So, rows with all zero elements, so if there

are rows, which are entirely 0, then they must lie at the bottom of the matrix, they

must come at the end, so they must be below those rows, which have a non-zero element.

So, these 3 are the requirements for a matrix to be in Row echelon form and then, if you

want the matrix to have one further property, which is the following, that for a non-zero

row, if you look at the leading entry, you look at that first one that you have and then

you look at the column in which the entire column. So, already the requirement that the second requirement that each leading entry in the column to the right, tells you that

below this 1 everything is 0, you cannot have anything non-zero below this 1. But you could

have non-zero entries above this 1. So, we call it, we say it is in the reduced Row echelon

form, if the entries above that leading 1 are 0.

So, the previous example that we had, which was 1 0 2, and 0 1 3, this is indeed in reduced

Row echelon form. So, first of all, it is in Row echelon form. Let us check that. So,

the first non-zero element of each row is a 1 indeed, that is the case because what

is the first non-zero element in each row, well this is the first non-zero element, this

is the first non-zero element and both of them are 1s.

And then the second requirement is that if you have a leading entry, you will look at

the column and all the leading entries in rows before that must be before it, so this

one over here this is in the second column, so if you look at the previous row, the 1 is in the first column. So, that is okay, that is exactly what we want. So, this is

satisfied and this is satisfied and rows with all zero elements, well, there is no such

row, so this is satisfied.

And now let us talk about reduced Row echelon form. So, it says, if you take these ones,

then the entire column other than that 1 is 0. Indeed, below this, the first 1 there is

a 0 and above the second 1, there is a 0, so that is also satisfied. So, we have seen

that in our previous example, indeed, satisfied this. So, it was not the reduced Row echelon

form and we could read off the solutions to Ax equals b very easily.

So, let us do another example. So, here is a matrix, which is in Row echelon form. So,

the first in each non-zero n row. So, all 3 rows are non-zero, the first non-zero term

is indeed 1 is that the case indeed, that is the case. 1, 1 and 1 and then whenever

you have a leading term, so you have this 1 over here, this is in the fourth column, so the 1s which are in the previous columns, so the 1 in the previous column, it is before

this it is in the third column. So, that matches up and then the 1 above this is in the first

column, so that matches up.

So, your 1s have to sort of go diagonally, but like this not, there cannot be 1 below

1, 1 to the below and left, it has to be below and right. So, this is in reduced echelon

form. So, let us ask, this is in Row echelon form? Is it in reduced Row echelon form? So,

for it to be in the reduced Row echelon form? For the leading terms, we have to look at

the columns and ask, are all the other entries 0? Well, that is certainly not the case over

here because here is a non-zero entry, here is a non-zero entry, here is a non-zero entry,

so this is not in reduced Row echelon form, this is in Row echelon form. So, let us look

at another example.

And we can do the same analysis for the second example. In fact, these matrices look very similar and there is a way of going from one to the other, which we are going to study

in the next video. So, let us look at this matrix over here. Is this in Row echelon form?

Well, certainly all the leading terms are 1, all the 1s which are leading terms come

to the right of the ones which are in the previous rows and there is no non-zero row,

so the third axiom is requirement is satisfied easily.

But this has a further property that not only is it in Row echelon form, it is in the reduced

Row echelon form, because now, these entries over here are indeed zeros and that is a requirement

for reduced row echelon form. So, this matrix is in row echelon form, this matrix, the first

one is in row echelon form. But it is not in reduced row echelon form

and this is in reduced row echelon form. And of course, if it is in reduced row echelon

form, it is already in row echelon form because reduced is a stronger requirement.

So, let us do the analysis that we did in the example that we saw earlier in this video, so let us ask what is the set of solutions for this? So, if we want, let us say Ax is

0, A r ref x is 0, what do we do, we write down again the system of equations. So, 1

plus 2 x 2 is 0, or maybe let us do b, let us do the general case, let us do b. So, b,

so this is b1 and then x3 is b2 and then x4 is b 3. So, now I want the set of solutions

for this. So, already we know what is x3 and x4, x3 and x4 are fixed they are b3 and b4

and for x1, what do I do? Well, I can do what we did before.

So, I can say x 1 is 1 minus 2 x2 and then if I choose a value for x2, automatically,

I can get a solution by substituting that value in this equation and getting and substituting

x1 to be that particular number. So, if I take x2 to be constant some c, then I can

get x1 to be b1 minus 2 c. So, this gives me a solution. So, what is the solution? The

solution is x is b1 minus 2c, c, b2, and b3, this is my solution for Ax is equal to b and

now whatever value of c you substitute, will give you a solution.

So, if you take c to be 0, for example, then your solution will be b1, 0, b2, b3. If you

take c to be 5, let us say then your solution will be b1 minus 10, 5, b2, b3. So, you can

see that I can read off solutions for these matrices in this way and not just some solutions,

we can in fact get read off all solutions. This is the only way we can get solutions

for these matrices and this is how, this is the use of the row echelon form or the reduced

row echelon form.

So, solutions of Ax is b when A is in reduced Row echelon form. We did not do one small

case in the examples, but I will state it here before I talk about the general situation.

So, let Ax is b be a system of linear equations and suppose A is in reduced row echelon form.

Suppose for some i, the i'th row of A is a 0 row, but bi is non-zero. So, the i'th row

of A is 0, but the corresponding constant bi is not 0, then this system has no solution.

You can think of for a second and if you still cannot see why then here is the reason. So, we can write down the corresponding system of equations and if you write down the system

of equations, what will the i th equation look like, the i th equation will look like

0 times x1 plus 0 times x2 plus 0 times xn because the entire i th row of A is 0, so

the constants, the coefficients there are 0.

But on the right-hand side, you have bi, so we want that 0 times x1 plus 0 times x2 plus

0 times x n is bi. Now the left-hand side, it does not matter what the xi s are, what

values you put inside xi s, the left-hand side will always evaluate to be 0. So, if

b is non-zero, then there is no hope of getting a solution. Whatever values of x1, x2, xn

and you take this equation cannot be satisfied.

So, there is no possible solution for this system of equations. So, the key point is

that there if row of A is 0, then the corresponding constant bi must be 0, in order for there

to be a solution, otherwise there cannot be a solution, there will not be a solution, we have seen why? So now, let us look at the case where such a thing does not happen. So,

whenever we have 0 rows in A then the corresponding constants bi are 0.

So, solutions again. So, suppose we have Ax is b and A is in the reduced Row echelon form.

So, assume that for every zero row of A, the corresponding entry of b is also 0. So, that

means if the i'th row of A is 0, then b is also 0, then we can apply the procedure that

we saw in the examples that we did before. So, we are going to give some names here,

so that we make this procedure very algorithmic.

So, if the i'th column has the leading entry of some row there are going to be some columns,

which have leading entries. In our first example of where we had 1 0 2, and 0 1 3, the leading

entries were in the first column and the second column respectively. So, then we will call

xi as a dependent variable. So, in that first example, since the leading entries were in

columns 1 and 2, x1 and x2 are going to be dependent variables.

And if the i'th column does not have the leading entry of some row, we call x i an independent

variable. So, all the other variables are independent variables. So, in that first example,

the independent variable is x3. In the second example, the independent variables where x2.

So, let us maybe write that down. So, we had 1, 0, 2, 0, 1, 3. So, here x1 and x2 are dependent

and x3 is independent. And in the second example, well, we had I think 1 2 0 0, and then 0 0

1 0, and then 0 0 0 1. So, here are the leading columns the leading entries are in columns

1, 3 and 4 respectively.

So, x1, x3, x4 are dependent variables and x2 is an independent variable and you saw

this reflected in the solutions, why we are calling it independent. So, if you remember

how we solve this in the first one, we took x3 to be, fixed x3 to be some arbitrary c

and then from there we got what is x1 and x2. And similarly, in the 2nd example, we

fixed x2 to be some arbitrary c and then x1, x3, and x4 for fixed, so that is what we are going to do.

Assign arbitrary values to the independent variables and then for a dependent variable,

what happens? There is a unique equation in which it occurs. So, if you look at the first

example, 1 0 2 and 0 1 3, x1 occurs exactly in 1 equation, it occurs in no other equation.

Similarly, for the second example, if you look at x1 it occurs in only 1 equation, x3

occurs and only 1 equation, x4 occurs in only 1 equation.

So, from those equations, we can find out their values. So, all other variables in that

equation are independent variables. That is why that is a power of the reduced row echelon

form. Because all the other column with a leading 1, the entries, all the other entries

are 0, so no dependent variables can occur together. In a single equation, you will have

only 1 dependent variable, if at all. So, all the other equations, variable in that

equation are independent variables and we have assigned values to them. So, from there, I can move them to the right-hand side and get the value of the dependent variable and

then if we put all these together the obtained values for xi give a solution to the system.

And, in fact, every solution can be obtained in this way, that is what we saw in the 2

examples that we did. So, the conclusion here is, if A is in the reduced row echelon form,

this very easy procedure, it is an algorithm. Let us recall what it is, you find out what

are the leading ones, the corresponding columns, the column numbers will tell you what are

the dependent variables, all the other ones are the independent variables.

For the independent variables, you assign arbitrary values and then calculate the values

of the dependent variables from the corresponding equations. So, this procedure provides us

with all the solutions of Ax is equals b. Thank you.